

It takes two: finding the normal modes in a coupled pendulum system

Brookelyn H Vermont

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

The normal modes of a system of two coupled pendulums were determined by using angular displacement data recorded with rotary motion sensors. As the pendulums oscillated, the angular displacements $\theta_1(t)$ and $\theta_2(t)$ were measured simultaneously and used to construct a covariance matrix describing their correlated motion. Diagonalization of this covariance matrix revealed two independent mode coordinates corresponding to in-phase and out-of-phase oscillations. Projection of the experimental data onto these mode coordinates produced a clear separation of the system's behaviors, with the in-phase mode dominating the motion.

I. INTRODUCTION

Synchronization is one of the most common phenomena observed in nature, yet the physical mechanisms underlying such behavior in non-linear systems are still an active area of study. Christiaan Huygens played a key role in the early mathematical understanding of pendular motion through his development of the first pendulum clock. While the motion of a single pendulum is well described by simple harmonic motion at small angles, many real physical systems often involve multiple oscillators that exchange energy. Huygens observed that when he hung two pendulums from a common support, it took about a half an hour for them to gradually synchronize, oscillating out of phase at the same frequency. He referred to this as “the sympathy of two clocks” [?]. When two pendulums are coupled in this manner, their behaviors become interdependent, leading to collective behavior that cannot be explained when analyzing each pendulum individually [?].

A defining feature of coupled pendulum systems is the existence of normal modes, which are specific patterns of motion in which the pendulums oscillate at a single characteristic frequency with a fixed phase relationship. For two coupled pendulums, these modes correspond to in-phase and out-of-phase oscillations, each with distinct frequency determined by the coupling strength and the physical parameters of the system [?]. By measuring the angular displacement of each pendulum as a function of time, the normal modes can be identified and analyzed. The coupled pendulum system thus serves as a foundational example for studying normal modes, synchronization, and energy transfer in more complex physical systems.

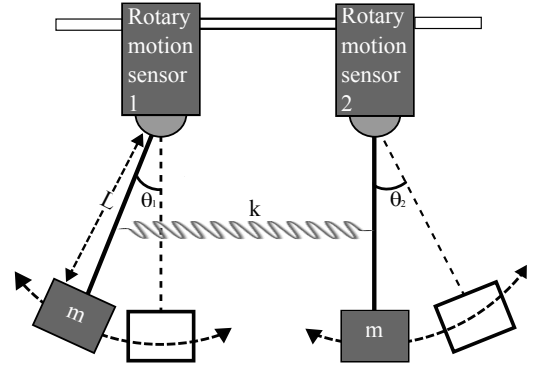


FIG. 1. A schematic of the coupled pendulums used in this experiment, consisting of two rotary motion sensors made up of massless rods with point masses at the bottom of each, the pendulums attached with a light spring.

II. KEY FINDINGS

This experiment consisted of two PASCO CI-6538 Rotary Motion Sensors mounted on a rigid point as seen in Fig.???. These sensors consisted of rods of negligible mass, each with a point mass fixed at the bottom of the rod. The two pendulums are connected in the center of the rods via a light spring. This connection allows for an energy transfer during oscillation. One pendulum was kept at the equilibrium position while the other was displaced by θ_1 and released from rest. Angular displacement was measured over 30-second intervals while the pendulums swung in beat. Data for ten runs were collected. Run 9 was chosen for the analysis because it had the greatest ratio of eigenvalues, indicating strong separation between the normal modes and minimal modal mixing.

The goal of the experiment is to identify the normal modes of the coupled pendulum. Normal modes are the characteristic patterns of motion in which the pendu-

lums oscillate at a single frequency with a fixed phase relationship. To do this, the measured angular displacements, $\theta_1(t)$ and $\theta_2(t)$, were first centered by subtracting the mean value for each pendulum. To quantify correlations between the two pendulums, the covariance matrix is constructed. Its elements are defined by

$$A_{11} = \langle (\theta_1 - \bar{\theta}_1)^2 \rangle, \quad (1)$$

$$A_{22} = \langle (\theta_2 - \bar{\theta}_2)^2 \rangle, \quad (2)$$

$$A_{12} = \langle (\theta_1 - \bar{\theta}_1)(\theta_2 - \bar{\theta}_2) \rangle, \quad (3)$$

where $\langle \cdot \rangle$ denotes a time average over the dataset.

The resulting covariance matrix is

$$\mathbf{A} = \begin{pmatrix} A_{11} & A_{12} \\ A_{12} & A_{22} \end{pmatrix}. \quad (4)$$

The normal modes of the coupled system correspond to the eigen vectors of the covariance matrix. The eigenvalues λ are obtained by solving the characteristic equation

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0, \quad (5)$$

which yields

$$\lambda_{1,2} = \frac{(A_{11} + A_{22}) \pm \sqrt{(A_{11} + A_{22})^2 - 4(A_{11}A_{22} - A_{12}^2)}}{2}. \quad (6)$$

The larger eigenvalue corresponds to the dominant mode of motion.

For each eigenvalue λ_k , the corresponding eigenvector $\mathbf{x}^{(k)}$ satisfies

$$(\mathbf{A} - \lambda_k \mathbf{I})\mathbf{x}^{(k)} = 0. \quad (7)$$

Solving this system yields eigenvectors of the form

$$\mathbf{x}^{(k)} = \begin{pmatrix} \frac{A_{12}}{\lambda_k - A_{11}} \\ 1 \end{pmatrix}, \quad (8)$$

which are normalized such that

$$\|\mathbf{x}^{(k)}\| = 1. \quad (9)$$

These eigenvectors represent the relative amplitudes of the two pendulums on each normal mode. The normal mode coordinates are obtained by projecting the angular displacement onto the eigenvectors:

$$\eta_1(t) = x_1^{(1)}\theta_1(t) + x_2^{(1)}\theta_2(t), \quad (10)$$

$$\eta_2(t) = x_1^{(2)}\theta_1(t) + x_2^{(2)}\theta_2(t). \quad (11)$$

The resulting time series $\eta_1(t)$ and $\eta_2(t)$ correspond to the in-phase and out-of-phase normal modes of the coupled pendulum system.

Using the angular displacement data from a representative experimental run, the covariance matrix was found to be

$$\mathbf{A} = \begin{pmatrix} 0.00944 & 0.00616 \\ 0.00616 & 0.01002 \end{pmatrix}, \quad (12)$$

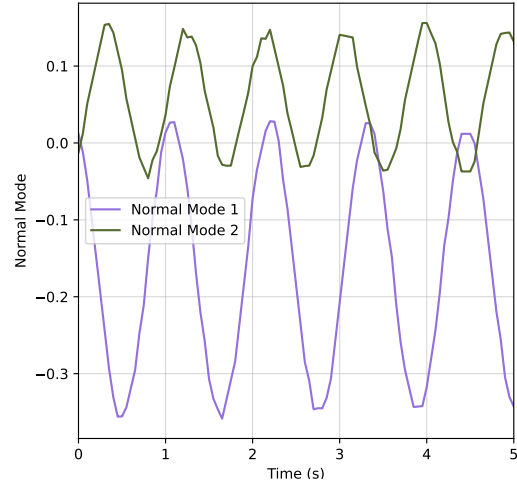


FIG. 2. Plot of normal modes 1&2 as a function of time. The in-phase mode (Normal Mode 1, bottom wave) dominates the motion, while the out-of-phase mode (Normal Mode 2, top wave) exhibits a smaller amplitude, consistent with the relative magnitudes of the covariance matrix eigenvalues.

where all entries are given in units of rad^2 .

Diagonalization of the covariance matrix yielded eigenvalues

$$\lambda_1 = 0.01590, \quad \lambda_2 = 0.00356, \quad (13)$$

with corresponding normalized eigenvectors

$$\mathbf{x}^{(1)} = \begin{pmatrix} 0.690 \\ 0.724 \end{pmatrix}, \quad \mathbf{x}^{(2)} = \begin{pmatrix} -0.724 \\ 0.690 \end{pmatrix}. \quad (14)$$

The normal mode coordinates were constructed by projecting the angular displacements onto the eigenvectors,

$$\eta_1(t) = 0.690\theta_1(t) + 0.724\theta_2(t), \quad (15)$$

$$\eta_2(t) = -0.724\theta_1(t) + 0.690\theta_2(t). \quad (16)$$

III. CONCLUSION

The first eigenvector has components of the same sign, indicating an in-phase mode in which both pendulums oscillate together. The second eigenvector has components of opposite sign, corresponding to an out-of-phase mode where the pendulums oscillate in opposition.

Fig.?? shows the time evolution of the two normal mode coordinates obtained from the angular displacement data. The motion is clearly decomposed into an in-phase component and an out-of-phase component, indicating that the coupled pendulum system is well described by two independent normal modes.

Normal Mode 1 exhibits a significantly larger amplitude than Normal Mode 2, consistent with the larger associated eigenvalue of the covariance matrix. This indicates that the majority of the system's variance, and

therefore the dominant motion, is contained in the in-phase mode. Physically, this corresponds to both pendulums oscillating together with similar amplitudes, suggesting that this mode is more easily excited by the initial conditions of the system.

Normal Mode 2 displays a smaller-amplitude oscillation and corresponds to out-of-phase motion of the pendulums. The reduced amplitude of this mode suggests weaker excitation, which may arise from asymmetries in the initial displacement or from dissipative effects that preferentially damp higher-energy relative motion between the pendulums.

The clean separation of the motion into two distinct mode coordinates demonstrates that the covariance matrix diagonalization effectively isolates the system's normal modes and confirms the expected behavior of a weakly coupled two-oscillator system.

The coupled pendulum system exhibits behaviors that are well captured by two independent normal modes corresponding to collective and relative motion. The successful extraction of these modes from experimental data demonstrates how normal mode behavior emerges naturally in real, imperfect systems and highlights covariance-based methods as a powerful tool for analyzing coupled oscillations.

While the covariance matrix method effectively isolates the normal modes, it is important to acknowledge sources of experimental error. Random fluctuations in the rotary sensor readings, finite sampling over 30-second intervals, and slight deviations from the ideal pendulum model introduce uncertainties in the measured eigenvalues and eigenvectors. Additionally, small damping effects may reduce the amplitude of the out-of-phase mode relative to the in-phase mode. Despite these limitations, the overall mode structure is clearly observed and consistent across multiple runs, indicating that the method is robust for identifying normal modes in real coupled systems.

Understanding the normal modes of coupled pendulums also has practical relevance in engineering and physics. For example, analyzing vibrations in bridges, mechanical structures, or coupled oscillatory circuits often relies on identifying independent modes to predict and control system behavior, illustrating the broader applicability of these experimental and analytical techniques.

ACKNOWLEDGMENTS

This research would not have been made possible without funding from the Carthage College Physics and Astronomy Department along with assistance from Dr. Joseph Anderson.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

Getting swingy with it: determining gravity using a simple pendulum

Brookelyn H. Vermont

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

This experiment uses a simple pendulum to measure the acceleration due to gravity. This is done by timing the pendulum's oscillations for several different string lengths. For small release angles, the motion of the pendulum can be approximated as simple harmonic motion, leading to the linear relationship between the period of oscillation and the square root of the length of the pendulum. The time for ten oscillations was measured for multiple trials for ten different lengths. A weighted linear fit of period versus the square root of length was used, whose slope was proportional to g . This gave $g = 10.01 \pm 0.04 \text{ m/s}^2$. This value is slightly larger than the accepted value, likely due to systematic timing uncertainties. The results clearly follow the expected linear relationship.

I. INTRODUCTION

Despite their apparent simplicity, pendulums have been important tools in physics for centuries, from Galileo's early observations of regular motion in swinging chandeliers to Foucault's use of a large pendulum to demonstrate Earth's rotation. Galileo's work on isochronism laid the foundation for the development of accurate timekeeping, and Christiaan Huygens later developed pendulum clocks, which greatly improved navigation and measurement.

A simple pendulum consists of a mass suspended from a fixed point by a light string. When displaced from equilibrium and released, the pendulum swings back and forth under the influence of gravity, with a restoring force directing it toward equilibrium. For small angular displacements, the motion is approximately periodic, and the period depends primarily on the length of the pendulum and the acceleration due to gravity [?]. Real pendulums deviate from the ideal due to air resistance, friction, and the finite mass of the string, providing an opportunity to explore the limitations of theoretical models and the role of experimental uncertainty.

In this experiment, the period of a simple pendulum was measured for several string lengths to investigate the relationship between oscillation period and pendulum length to determine the acceleration due to gravity. By analyzing how the period changes with length, this study demonstrates the principles of simple harmonic motion while highlighting experimental limitations. Even with simple equipment, the experiment illustrates how classical mechanics can be connected to hands-on observations, making it accessible for both educational and research purposes.

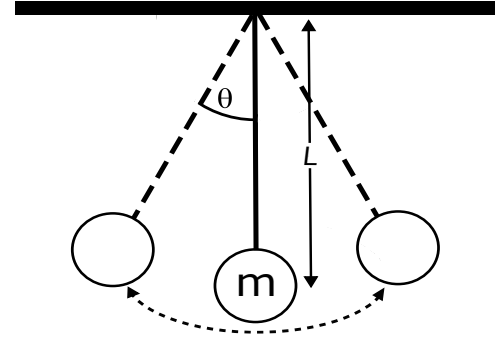


FIG. 1. Diagram of experimental setup, showing simple pendulum with mass m , length L , and small angle θ .

II. KEY FINDINGS

A simple pendulum is a system consisting of a point mass m suspended from a fixed pivot by a massless, inextensible string of length L . When displaced from its equilibrium position and then released, the pendulum swings back and forth due to the force of gravity. The force of gravity acts vertically on the mass, and only the component of this force along the arc of motion contributes to restoring the pendulum toward its equilibrium. This component of the gravitational force creates a restoring torque about the pivot that returns the pendulum to its lowest point.

The restoring torque τ is related to the angular displacement θ from equilibrium:

$$\tau = -mgL \sin \theta, \quad (1)$$

where the negative sign indicates that the torque acts against the displacement. The torque generates angular acceleration α according to Newton's second law for rotation:

$$\tau = I\alpha, \quad (2)$$

where $I = mL^2$ is the moment of inertia of the mass about the pivot, and $\alpha = d^2\theta/dt^2$ is the angular acceleration. Therefore, the angular displacement of the pendulum satisfies the nonlinear equation:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0. \quad (3)$$

For small angular displacements, ($\theta \leq 15^\circ$), $\sin \theta \approx \theta$. This small-angle approximation linearizes the equation, making it mathematically equivalent to a simple harmonic oscillator. Physically, this means the restoring torque is approximately proportional to the displacement, which results in periodic motion with a fixed frequency. The resulting linear differential equation is:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \theta = 0, \quad (4)$$

a second-order ODE whose solution has two constants of integration representing the initial amplitude and phase. The angular frequency is

$$\omega = \sqrt{\frac{g}{L}}. \quad (5)$$

The period of oscillation, $T = 2\pi/\omega$, is now

$$T = 2\pi\sqrt{\frac{L}{g}}. \quad (6)$$

This predicts a linear relationship between T and $\sqrt{L}$.

The experimental apparatus shown schematically in Fig. ?? consisted of a simple pendulum formed by a spherical metal mass suspended from a fixed support by an inextensible string of negligible mass. The string was attached to a fixed rod supported by the edge of a rigid horizontal surface, forming a fixed pivot point about which the pendulum oscillated. The pendulum length L was defined as the distance from the pivot point to the center of mass of the metal sphere.

Oscillations began with a small angular displacement $\theta = 10^\circ - 15^\circ$ to satisfy the small-angle approximation. Three different string lengths were used while the mass of the pendulum was held constant. For each length, the time for ten complete oscillations was measured using a handheld stopwatch. Each measurement was repeated three times and averaged to reduce random timing uncertainty.

The plot of measured values for the period versus the square root of the pendulum length is shown in Fig. ?. A weighted linear regression was performed, forcing the fit through the origin, consistent with the theoretical relationship $T \propto \sqrt{L}$. The slope of this line, M , represents a ratio of 2π to g . Rearranging this ratio gives the calculated value of gravitational acceleration:

$$g = 10.07(2M)^2 \pm 0.04 \text{ m/s}^2. \quad (7)$$

The accepted value of gravitational acceleration is

$$g_{\text{accepted}} = 9.80665 \text{ m/s}^2, \quad (8)$$

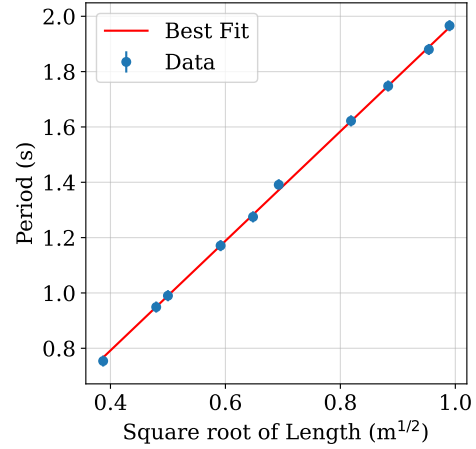


FIG. 2. The average of the measured periods is plotted against the square root of the measured lengths. The solid line is a least squares regression fit.

as reported in [?]. The experimentally determined value differs from the accepted value by 2.6%.

Despite this small discrepancy, the data clearly demonstrates the expected linear dependence of period on the square root of length. The slight overestimation of g likely arises from systematic effects. The dominating systematic error is reaction time bias in timing, estimated at ± 0.02 seconds per period, or 0.2 seconds for the time of $N=10$ oscillations. Random variations due to human reaction time were minimized by averaging over multiple oscillations for each length.

III. CONCLUSION

In this experiment, the period of a simple pendulum was measured for multiple string lengths to determine the acceleration due to gravity. The results show a clear linear relationship between the period and the square root of the pendulum length, consistent with the theoretical model for small-angle oscillations. The experimentally determined value, $g = 10.07 \pm 0.04 \text{ m/s}^2$, shows a small overestimation relative to the accepted value. This is primarily due to systematic factors such as reaction time bias, finite release angles, and uncertainties in effective pendulum length.

The experiment is subject to both systematic and random errors that affect the accuracy of the measured gravitational acceleration. While statistical uncertainty from the regression is minimal, systematic errors dominate the overall uncertainty. Systematic errors include reaction time bias in timing, deviations from the small-angle approximation, minor elasticity in the string, and potential misalignment of the pivot, all of which can slightly increase or decrease the measured period. Human reaction time bias when timing was the dominate error factor in

the experiment, due to changes from trial to trial. During the data analysis the time error was weighted while the length measurement or angle error was considered negligible.

To reduce these errors in future experiments, several improvements could be implemented. Using automated timing systems such as photogates would eliminate reaction-time uncertainty, while increasing the number of oscillations per measurement would reduce the relative effect of timing variability. Additionally, taking more measurements at a wider range of pendulum lengths would improve the linear fit and decrease the statistical uncertainty in the slope used to calculate g . Careful control of the release angle and using a more rigid pivot and inextensible string would further minimize systematic deviations from the ideal simple pendulum model.

Overall, the experiment successfully illustrates fundamental principles of classical mechanics, demonstrating how a simple pendulum can be used to determine acceleration due to gravity. With improved measurement techniques, this system offers both educational and research value in exploring harmonic motion and the limitations of idealized models.

ACKNOWLEDGMENTS

This work was assisted by lab partner Braedon Larsen. Additionally, this research would not have been made possible without funding from the Carthage College Physics and Astronomy Department along with assistance from Dr. Joseph Anderson.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

Riding the wave: determining laser wavelength using a Michelson interferometer

Brookelyn H. Vermont

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

The wavelength of a helium-neon gas laser was determined using a Michelson interferometer by relating mirror displacement to interference fringe shifts. As the movable mirror was displaced, the number of fringe shifts m was counted and the corresponding mirror displacement d was recorded. Plotting d as a function of m revealed a linear relationship, where the slope corresponds to half the wavelength of the light, $d/m = \lambda/2$. Using this result, the wavelength was determined to be $\lambda = 653(6)\text{nm}$.

I. INTRODUCTION

Michelson interferometers have been key components in many scientific applications, ranging from optical calibration on an optics table, to measuring the diameter of stars in stellar interferometry. The behavior of light was heavily debated with Isaac Newton proposing that light consists of many minute particles, while Christiaan Huygens argued that light behaved as a wave [?]. This debate persisted until Thomas Young's double-slit experiment provided strong evidence for the wave nature of light through the observation of interference patterns [?].

Interference patterns were crucial in proving light as a wave phenomenon. When two coherent light beams overlap, their relative phase determines whether they interfere constructively or destructively, producing a pattern of dark and bright fringes. In a Michelson interferometer this relative phase depends on the optical path difference between the two beams. By moving one of the mirrors the path difference changes in a controlled manner, causing the interference fringes to shift. By counting the number of fringes that pass a fixed reference point as the mirror is displaced, the wavelength of the light can be determined. The Michelson interferometer was developed by Albert A. Michelson to measure small differences in optical path length with extreme precision [?]. Michelson interferometry remains a foundational technique in modern physics, with applications ranging from precision metrology to gravitational wave detection.

II. KEY FINDINGS

The relationship between mirror displacement and fringe shifts stems from classical wave optics, where the phase of a light depends on its optical path length. In a Michelson interferometer (Fig. ??), light from a laser is divided by a beam-splitter into two beams that propagate along perpendicular arms. One beam travels to a

fixed mirror, while the other travels to a movable mirror. The reflected beams return to the beam-splitter, where they recombine and produce an interference pattern at a viewing screen. When one mirror in the interferometer is moved by a distance d , the optical path length changes by $2d$, since the beam travels to the mirror and back. This change in path length alters the relative phase between the two beams, creating the predictable fringe pattern.

Each time the optical path length changes by one wavelength one fringe shift is observed. Measuring the mirror displacement over a known number of fringes therefore yields a linear relationship:

$$\lambda = \frac{2d}{m}, \quad (1)$$

where λ is the wavelength, d is the mirror displacement, and m is the number of observed fringes.

The experiment used a PASCO OS-8514 helium-neon gas laser and a PASCO scientific Model OS-8501 interferometer consisting of a beam-splitter, a stationary mirror (M1), a moveable mirror (M2), and a viewing screen as shown in Fig.???. The laser was mounted on an optics bench, with the interferometer positioned approximately 20 centimeters away. The beam-splitter was oriented at 45° to direct light toward the stationary mirror, while the moveable mirror faced the laser. The viewing screen was placed behind the beam-splitter to observe the interference pattern.

After aligning the interferometer to produce a clear circular fringe pattern as shown in Fig.???, data was collected by counting a predetermined number of fringes while measuring the corresponding displacement of mirror M2 using a micrometer. A fringe shift occurs when the optical path length of one arm changes, by moving M2, forcing the interference pattern to rearrange. The circular pattern in Fig.???. is due to M1 and M2 being parallel, and a shift causes the rings to expand or contract. A single fringe shift is counted from the center of one dark ring to the next. An error for the distance measured was estimated to be $\delta = 0.5\mu\text{m} = 500\text{nm}$ because the micrometer was marked in increments of one

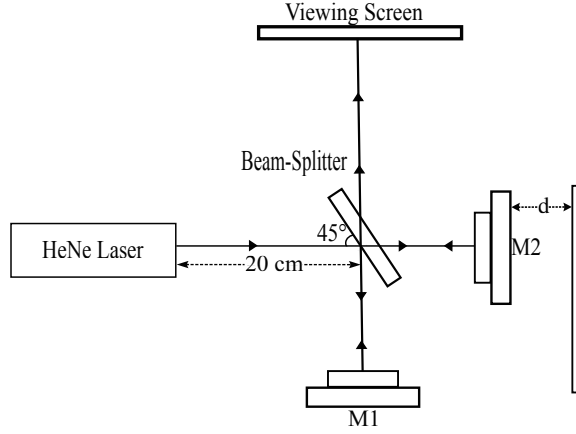


FIG. 1. A schematic of the Michelson interferometer used in this experiment. Light from a helium–neon laser is split by a beam-splitter and reflected by a stationary mirror (M1) and a movable mirror (M2). The two beams recombine at the beam splitter and form an interference pattern on the viewing screen. Displacement, d , of mirror M2 changes the optical path difference between the beams.

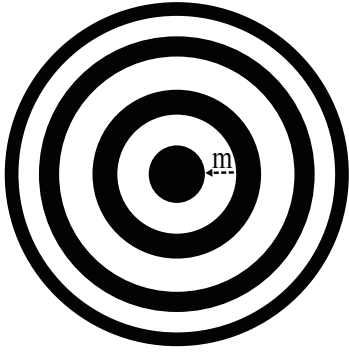


FIG. 2. Interference fringe pattern observed on the viewing screen after alignment of the Michelson interferometer. Bright and dark circular fringes result from constructive and destructive interference of the two coherent beams. Fringe shifts, m , were counted as mirror M2 was displaced.

micrometer, making it hard to exactly distinguish an exact measurement. A mechanical band ensured that M2 remained parallel with M1 while being moved. Measurements began with 20 fringes and increased in increments of 5 up to 70 fringes. Measuring larger numbers of fringes reduced the relative uncertainty in the wavelength calculation.

The wavelength of the helium-neon laser was determined by measuring the mirror displacement required to produce a known number of interference fringe shifts. Mirror displacement was plotted as a function of fringe count as seen in Fig. 3. There is a clear linear relationship between the two, indicating that the mirror displacement increased proportionally with the number of observed fringes. The linear relationship observed is expected from the theoretical prediction that the mir-

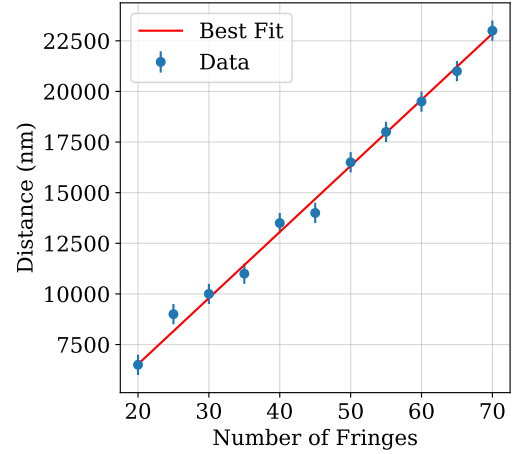


FIG. 3. Mirror displacement as a function of fringe count for a HeNe laser. A least squares linear fit with the y-intercept constrained to zero was applied to the data, yielding a slope of $M = 326 \pm 3$ nm/fringe.

ror displacement is proportional to the number of fringe shifts. In Fig. ??, the y-intercept is constrained to the origin, $(0, 0)$, due to the experimental setup in which distance is measured as a function of the number of fringes. Therefore, if zero fringe shifts have occurred, the measured distance is zero.

A linear fit to the data yielded a slope corresponding to half the wavelength of the light source. Using this slope, the wavelength of the laser was determined to be

$$\lambda = \frac{2d}{m} = 653(6) \text{ nm.} \quad (2)$$

The primary source of uncertainty stemmed from the measured distance of the mirror displacement. Because fringes are discrete and easily counted, uncertainty in fringe number was negligible compared to uncertainty in the measured mirror displacement. The measured wavelength was consistent with the true wavelength value of a helium-neon gas laser of $\lambda = 632.8$ nm [?].

III. CONCLUSION

The wavelength of the helium-neon laser was successfully determined using a Michelson interferometer by relating mirror displacements to interference fringe shifts. The measured value corresponds to a z-score of 3.19 when compared to the expected wavelength, which gives a two-tailed p-value of approximately 0.0014. This indicates that the measured wavelength differs from the expected value by over three standard deviations, making the difference statistically significant. The accuracy of the measurement was primarily limited by uncertainty in the mirror displacement. The micrometer was marked in increments of one micrometer, making it hard to exactly dis-

tinguish an exact measurement between a $0.5\text{ }\mu\text{m}$ increment. This difficulty in exact measurement is how the error for distance measured was estimated to be $\delta = 0.5\text{ }\mu\text{m}$ $\bar{5}00\text{ nm}$. Future measurements could reduce these effects by using automated fringe detection or larger mirror displacements.

ACKNOWLEDGMENTS

This research would not have been made possible without funding from the Carthage College Physics and Astronomy Department along with assistance from Dr. Joseph Anderson.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

The Charged Family: Verifying and quantizing electrons using electromagnetic forces

Braedon Larsen

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

When exposed to a magnetic field, charged particles such as electrons experience a force. If that particle is also moving at the time that force becomes a force known as a Lorentz force, which for the case of a magnetic field generated by a Helmholtz coil, will cause electrons to engage in cyclotron motion. Determining the radius of this motion and knowing the velocity and magnetic field strength can allow the charge to mass ratio to be determined. As the average radius of motion and the square root of accelerating voltage over the magnetic field strength has a linear relationship that is proportional to the ratio of electron charge and mass. The determined ratio was found to be $\text{Ratio} = (1.004 \pm .2509) \times 10^{11} \text{ C/kg}$.

I. INTRODUCTION

Electrons are a fundamental particle that orbits the nuclei of atoms. They are known in the modern day for carrying the energy that we see as electricity through metals, and on a fundamental level having a negative charge, and being very light compared to protons or neutrons. However, not long ago, electrons weren't even known to exist, instead their effects were observed through their interactions with matter (again, electricity). The discovery of the electron as a discrete particle occurred in 1897 by J.J Thomson who was studying how electricity moves through cathode ray tubes[?]. Specifically, Thomson was using EM-fields inside a cathode ray tube to manipulate the path of the what we now know as an electron beam. A cathode ray tube consists of a metal filament that is heated in a low pressure environment. When heated sufficiently, the filament would release a beam of electrons dependent on the temperature and type of material in an effect known as thermionic emission [?]. The EM-fields cause deflections in the path of the electrons, which Thomson measured and through several experiments was able to quantify the ratio of charge to mass of the members of this beam [?]. From these results, Thomson was able to conclude that the beam had to be made up of small discrete particles which he called "corpuscles." This was because the ratio was constant and much larger than other materials [?]. The physical results that Thomson discovered paved the way for many advancements in the fields of nuclear physics and the discovery of the first sub-atomic particle. Because of its significance, this ratio is found today in many experiments to show the physical characteristics of electrons. This experiment seeks to replicate Thomson's results, with perhaps more modern equipment. Instead of a cathode tube of different materials, an electron gun in a low pressure vessel will be used as an electron source, and a Helmholtz

coil with a constant magnetic field as a source of deflection.

II. METHODS AND RESULTS

Quantifying these physical results can be easily understood using some basic mathematical principles. It is known that any particle with charge traveling with respect to a EM-field, will have experience a mutually orthogonal force to the direction of travel and EM-field. This force is known as the Lorentz Force, for the case the traveling charge is exposed to no electric field once it has reached its starting velocity. Therefore, the Lorentz force can be described by the equation

$$\mathbf{F}_L = |q_e|\mathbf{v}\mathbf{B} \quad (1)$$

where q_e is the charge of the particle, $\mathbf{B}$ is the magnetic field, and $\mathbf{v}$ is the velocity of the particle. In this experiment, the uniform magnetic field will always be orthogonal to the direction of travel and will therefore cause the force to act centripetally. This allows the simplification of the expression to only the magnitude and using the form of centripetal force. This leads to the simplification of (??) to be the magnitudes of the vectors always pointing in the centripetal (mutually orthogonal to $\mathbf{v}$ and $\mathbf{B}$ direction.

$$F_L = |q_e|vB \quad (2)$$

Where F_L is the magnitude of the Lorentz force, q_e is the charge of an electron, v is the magnitude of an electron's velocity, and B is the magnitude of the magnetic field strength.

The other simplification that comes from the Lorentz force acting centripetally, is that the magnitude of the

force is equivalent to a centripetal force, therefore:

$$F_L = |q_e|vB = \frac{m_e v^2}{r} \quad (3)$$

$$|q_e|B = \frac{m_e v}{r} \quad (4)$$

For this experiment, an electron gun applies a voltage which electrons being charged particles accelerate across. This process gives the particles an amount of kinetic energy proportional to the voltage and the magnitude of the charge of an electron. This amount of kinetic energy is also proportional to the velocity, and combining the two expressions can yield: v

$$|q_e|V_{acc} = \frac{1}{2}m_e v^2 \quad (5)$$

$$v = \sqrt{\frac{2|q_e|V}{m_e}} \quad (6)$$

where r is the radius of the centripetal motion that the electron undergoes. V_{acc} is the accelerating voltage, and m_e is the mass of an electron.

Substituting in for the factors of velocity in the Lorentz force, a final function for the resulting centripetal radius as a function of magnetic field strength and accelerating voltage can be found.

$$r(B, V_{acc}) = \frac{\sqrt{V_{acc}}}{B} \sqrt{\frac{m_e}{|q_e|}} \quad (7)$$

The experiment performed was as discussed, a refined version of the ones performed by J.J Thomson. The apparatus used is a Uchida TG-13 with a diagram in ?? . It consists of a Helmholtz coil, which generates a uniform magnetic field within the bounds of its coil, this provides a magnetic field that varies with the current applied to the coil to deflect with. The apparatus includes a coefficient to convert from the applied current to the magnetic field strength ($B = C_B I$), with the coefficient having a value of $C_B = 4.804 \times 10^{-4} \frac{wb}{m^2}$. This means that (??) can be expressed in terms of current on the Helmholtz coils by the expression:

$$r(I, V_{acc}) = \frac{\sqrt{V_{acc}}}{IC_B} \sqrt{\frac{m_e}{|q_e|}} \quad (8)$$

with C_B being the magnetic coefficient from the TG-13. However for purposes of data analysis and visualization, the equation will be expressed in terms of the magnetic field strength (B). The electron gun is the accelerating component placed at the bottom of a low-pressure glass bulb. Once the excited electrons are engaged in centripetal motion, the radius of the resulting circle can be observed with a ruler attached to the back of the Helmholtz coils. Combining these measurements of radius, coil current, and accelerating voltage, (??) can be used to develop a linear relationship between the values. This relationship can be leveraged to solve for the desired ratio of $\frac{|q_e|}{m_e}$.

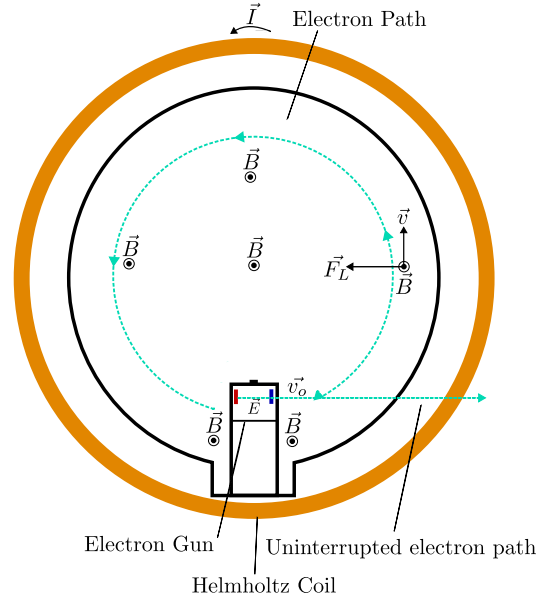


FIG. 1. Diagram of the Uchida TG-13 experimental apparatus, with the key components for the experiment, and the uninterrupted and actual electron path noted. In addition the vector directions of key quantities are also present on the diagram

V_{acc} (V)	I (mA)	Left Radius (cm)	Right Radius (cm)
219	13.1	$4.6 \pm .5$	$4.9 \pm .2$
219	16.7	$4.6 \pm .5$	$3.7 \pm .2$
219	14.0	$4.5 \pm .5$	$4.4 \pm .2$
259	15.3	$4.6 \pm .5$	$4.4 \pm .2$
259	17.2	$4.1 \pm .5$	$3.9 \pm .2$
281	14.8	$4.7 \pm .5$	$4.7 \pm .2$
281	17.2	$4.4 \pm .5$	$4.2 \pm .2$
296	14.8	$4.7 \pm .5$	$4.9 \pm .2$
296	17.2	$4.4 \pm .5$	$4.3 \pm .2$
296	22.2	$3.4 \pm .5$	$3.2 \pm .2$
256	22.2	$2.9 \pm .5$	$2.9 \pm .2$
224	22.5	$2.8 \pm .5$	$2.7 \pm .2$

TABLE I. Compiled experimental data consisting of: electron gun voltage, Helmholtz Coil current, and observed radius at two points, left and right with uncertainty

Experimentation was performed for different electron gun voltage, and at each voltage multiple currents were used, and the resulting radii recorded. The plot shown below in ?? is the average observed radius over the ratio of the square root of voltage over the magnetic field strength, the linear regression least squares fit, and the propagated error to the fit of the least squares fit intercepting at the weighted mean of the data set.

From these measurements, data analysis was performed [?] by using the relation in (??) with the linear regression line of best fit that was produced. Using this relation it could be found that the slope of the best fit line was equivalent to $\sqrt{\frac{m_e}{q_e}}$. Using this relation, combined with propagating the error through the slope and

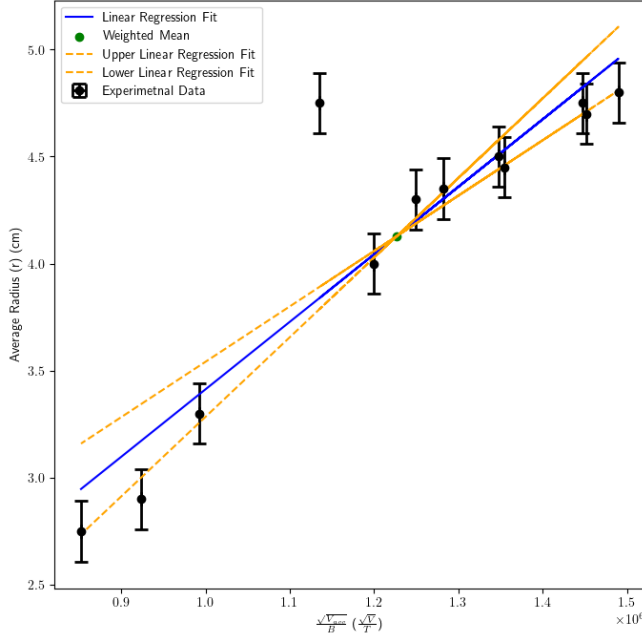


FIG. 2. r vs $\frac{\sqrt{V}}{B}$ plot with the least linear regression best fit line and the associated error to the slope based on the propagation of error from the initial measurements. The slope of the least squares fit line is $m = 3.16 \times 10^{-6} \pm 9.86 \times 10^{-7} \frac{\text{cm}}{\frac{\sqrt{V}}{T}}$ and a portrayed intercept of $b = .257$, however as seen in the supporting documentation [?] the intercept is taken to be 0 for its physical meaning and ignored otherwise.

onto the final value. The magnitude for this value for this quantity was found to be $\text{Ratio} = (1.004 \pm .2509) \times 10^{11} \text{ C/kg}$. Based on more precise measurements taken by NIST, an accepted value for this quantity is taken to be $\text{NIST Value} = 1.758 \times 10^{11} \pm 3.1 \times 10^{-10} \text{ C/kg}$. This places the experimentally found value to be within 57% of the NIST value. While the relative uncertainty of the measured value is 43.6%, this suggests that the calculated value was somewhat close to the reference value. The remaining difference could suggest some form of systemic error to the experiment, such as defects in the equipment used or interference from the environment affecting the path of the electrons.

III. CONCLUSIONS

In summary, the use of electric and magnetic fields, can be used to guide and control electrons. In this case, electrons were accelerated from an electron gun by a constant voltage (V_{acc}) in a low pressure environment. Once accelerated, they were subject to a constant magnetic field orthogonal to their path of travel, produced by a Helmholtz Coil with magnetic field strength (B) running through it. This magnetic field caused the electrons to orbit around a center point in centripetal motion, which provided a measurable radius, from which, the ratio of

charge to mass was found. That ratio was found to be $\text{Ratio} = (1.004 \pm .2509) \times 10^{11} \text{ C/kg}$. Within a serviceable range to the reference value of NIST Value = $1.758 \times 10^{11} \pm 3.1 \times 10^{-10} \text{ C/kg}$. This ratio was determined by fitting a least linear squares regression to the collected data as graphed in Fig. ?? and using (??). Future experiments making the same measurement would benefit from several improvements to improve the accuracy of the data. One major improvement would be a more precise way to measure the radius of the electron path, the current method is prone to bias based on the measurer. A better method would be to use some reference to track the path of the electron, such as a checkerboard and see where it intercepts instead of attempting to align a reflection, this could allow for more reproducible and consistent results. Finally, a way to reduce error in the system is increasing the number of points at which each circle is measured, since with more measurements the average uncertainty falls to a point.

ACKNOWLEDGMENTS

The author would like to acknowledge Brookelyn Velmont for her assistance with the data collection involved for this experiment.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

When ionized oil proved quantization: Data analysis techniques in ionized drops of oil for determining electron charge

Braedon Larsen

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

The charge on an electron is also referred to as the elementary charge e and is the quantum for the charge on a particle. Because of the quantization, charge on a particle will be an integer multiple of the elementary charge. By recording the velocities of ionized oil drops when naturally falling and when under the influence of a known electric field, then normalizing these charges, their existence as integer multiples of a single value can be seen. This value is the elementary charge and through the analysis done in this report was found to be $e = (1.569 \pm .0913) \times 10^{-19}\text{C}$.

I. INTRODUCTION

Charge is a fundamental property of matter that governs how that matter interacts with electric forces such as electric or magnetic fields. Greater charges lead to increasing effects that EM-fields have on any particular particle. In 1897, physicist J.J Thomson had discovered the existence of a charged particle that is now known as the electron[?]. Thomson had also quantified the ratio of the charge of this electron to its mass, but no measurement of its charge or its mass existed yet. However, about a decade after Thomson's initial measurement, there were many scientists taking different approaches in attempting to quantify either the mass or the charge of the electron. The first to succeed was Robert Millikan who quantified the charge of an electron, and since the ratio of the electron's charge to mass was known, he also quantified the mass of an electron. His experiment, now known as the "Millikan Oil Drop Experiment" took a different approach to many others, relying on fluid mechanics and the net forces exerted on a particle. Millikan used small drops of oil, so small that the viscosity of air made them fall at a constant observable rate. Using a source of ionizing radiation, a random net charge is applied to the oil drop. If these charges can be determined for a statistically sufficient number of oil drops, the differences between the charges could be taken, from these differences, a pattern emerged where the differences were all integer multiples of a minimum charge difference. This result verified the theory that charge was a quantized value and that quanta was the charge of an electron. In one experiment Millikan verified the charge of an electron which is now called e or the elementary charge, and that the elementary charge is the smallest unit of charge that is possible.

For this report, the experiment Millikan performed will be replicated and analyzed using modern data analysis techniques for both verifying the value of the elementary charge and the nature of charge as a quantized quantity.

II. METHODOLOGY AND RESULTS

For experimentation, a AP-8210 [?], was used with Squibb#5597 as the mineral oil [?], and Thorium-232 [?] as the source of ionizing radiation. After calibration of the apparatus, multiple attempts for data collection were performed.

The AP-8210 consisted of a small chamber enclosed on all sides but for a small hole in the cap, the upper and lower walls of the chamber were the plates of a capacitor, and the sides a plastic spacer. The hole was used in combination with a atomizer to spray small drops of oil into the chamber that could then be briefly exposed to the ionizing radiation before testing. The drops were viewed through a 30x magnification scope with a reticle marking every 0.5 mm and 0.1 mm. For each of these attempts, oil drops were introduced to the viewing chamber and ionized, the drops were then evaluated to select a drop that both: reacted to the polarities of the capacitor, and naturally fall at an appreciable rate. After a proper drop was selected, the time for the drop to travel 0.5 mm was measured in 3 configurations: falling naturally, rising with positive top plate polarity, and falling with negative top plate polarity.

Between all runs 3 drops were found to meet these criteria, and for these 3 drops, 6 ionization states were achieved. The additional states were achieved by re-ionizing the same drop after the first runs of data were collected. For these re-ionizations, the natural falling velocity was unchanged as the drop had nearly the same mass (any mass loss or gain due to gaining or losing charged particles is negligible).

To find the charge in a oil drop, we begin with free body diagrams for the potential states that the drop can be in. The drop has a net charge q , which will be affected by an electrostatic force when there is polarity on the plates. As can be seen in ??, the two cases are with field polarity or no field polarity. The two cases are considered to enter equilibrium instantaneously on the timescales used, the

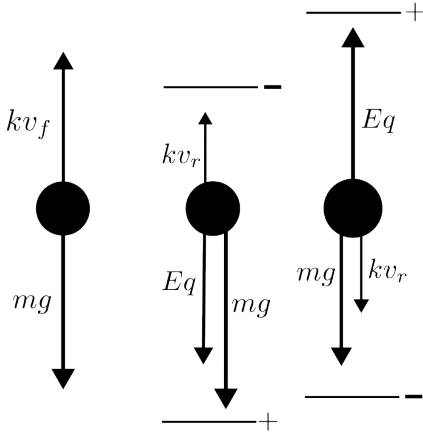


FIG. 1. Free body diagrams of the 3 potential states for the oil drop. Leftmost is the neutral state with the only forces being gravity and drag, the middle is the top plate negative state where the electric field causes a downwards force on the drop, and the right is the top plate positive state causing the drop to move upwards against gravity

forces will balance and create the following equations

$$\text{For no field polarity: } mg = kv_f \quad (1)$$

$$\text{For top plate + charged field: } Eq = mg + kv_r \quad (2)$$

$$\text{For top plate - charged field: } Eq + mg = kv_r \quad (3)$$

Where m is the mass of the droplet, g is acceleration due to gravity, v_f is the terminal velocity when the drop naturally falls, v_r is the terminal velocity when the drop is under the influence of the electric field, k is the coefficient of drag, and E is the electric field strength. Solving (??) and (??) for q yields the following:

$$q = \frac{mg(v_f + v_r)}{Ev_f} \quad (4)$$

The mass of the drop can be expanded using the density ρ of the oil

$$m = \frac{4}{3}\pi a^3 \rho \quad (5)$$

Where ρ is the density of the oil, and a is the radius of the drop. The radius of the drop a can be further expanded using Stoke's Law relating the radius of a sphere and its velocity when falling in a viscous medium.

$$a = \sqrt{\frac{b^2}{2P} + \frac{9\eta v_f}{2g\rho} - \frac{b}{2P}} \quad (6)$$

Where b is a constant equal to $b = 8.2 \times 10^{-3} \text{ Pa} \cdot \text{m}$, P is the pressure of the environment, and η is the viscosity of air.

TABLE I. Experimental constants

$\rho_{\text{oil}} = 886 \text{ kg/m}^3$	$b = 8.20 \times 10^{-3} \text{ Pa} \cdot \text{m}$
$g = 9.81 \text{ m/s}^2$	$P = 101325 \text{ Pa}$
$\eta_{\text{air}} = 1.848 \times 10^{-5} \text{ N} \cdot \text{s/m}^2$	$d = 7.560 \text{ mm}$
(at 25.5°C)	$V = 500.8 \text{ V}$

This term for a can be substituted into (??) and simplified to the final form of:

$$q = \frac{4}{3}\pi\rho g \left[\sqrt{\frac{b^2}{2P} + \frac{9\eta v_f}{2g\rho}} - \frac{b}{2P} \right]^3 \frac{v_f + v_r}{Ev_f} \quad (7)$$

After collecting the data, several processes were used in the data analysis. This process will be covered briefly here but is explained in greater detail in the supplemental information [?]. For each of the charge states, an averaged falling velocity $\bar{v}_f$, an averaged positive rising velocity $\bar{v}_+$, and an averaged negative rising (falling) velocity $\bar{v}_-$. With these velocities for each of the 6 charge states, they were then used with (??) to calculate a net charge for each charge state for the positive and negative cases. The values of the constants used in the calculations can be found below in ??

These positive and negative charges were averaged to get a single charge value for each charge state q_i . These charge states were then normalized by the smallest charge state, then the normalized charges were multiplied by the integer that yielded a minimum total distance of each normalized charge from its nearest integer neighbor. This is because as charge is quantized, the values for charge on a particle should follow a integer multiple of the fundamental charge, this method works to find what the integer multiples are in the data. To verify the distance to the nearest integer, a trial and error method was used since there was so few integers to check (1-11). The ideal integer, was selected by which integer brought the dataset closest to a least squares fit line. The results of this fitting can be seen in ?? and the final linear equation found whose slope is equivalent to the fundamental charge.

From ??, a refined estimate for the value of e is obtained as the slope of the linear least squares fit on that plot. This value was found to be $e = (1.569 \pm .0913) \times 10^{-19} \text{ C}$. Compared to the accepted value from NIST ($e = 1.602 \times 10^{-19} \text{ C}$) [?], there is a z-score of .357 and the best value itself is within 1.9% of the accepted value. The z-score value implies that the range calculated is within .357 standard deviations of the dataset to the mean or in this case accepted value being an underestimate by the positive sign for the z-score. It is clear in this case that the methodology of the experiment performed properly.

III. CONCLUSIONS

In summary, the use of droplets small enough that air behaves viscously toward them allows their measured ve-

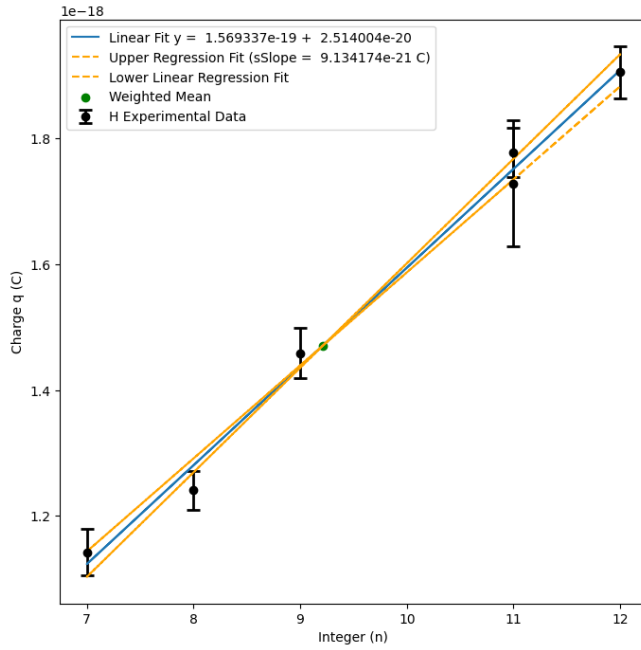


FIG. 2. Charge vs n -integer plot, used to find the best estimate for the charge of an electron e . The linear best fit line's slope is equivalent to e , with the linear best fit following the linear equation $y = (1.569 \pm .0913) \times 10^{-19}x + 1.149 \times 10^{-20}$. In addition the plot shows the points of experimental data, the weighted mean, and a visualization of the error to the slope.

locities to be used to calculate the charge on the particles when ionized to react to an electric field, and by consequence of charge quantization, calculate the fundamental charge e . After collecting a data set of velocities, a several step data analysis process was performed to go from the collected drop velocities to the charge on each drop. Once the charge for each drop was obtained, the data was fitted using the theory that charge was quantized and the charge values would be integer multiples of the fundamental charge. This analysis lead to a final value for the fundamental charge of $e = (1.569 \pm .0913) \times 10^{-19} \text{C}$. This value is consistent with the accepted value. For a greater certainty to the value, more data points per charged state could be used. Using a larger dataset would provide more evidence for the quantization of charge by showing the smallest difference that charged particles could have between their charges.

ACKNOWLEDGMENTS

The author would like to acknowledge Sema'je Farmer for his assistance with the data collection involved for this experiment.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

Determining the interplanar spacing of graphite using electron diffraction

Sema'Je Zechariah Farmer

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

When firing a beam of electrons at a crystal lattice, those electrons will diffract. The pattern that forms as a result of that diffraction will depend on and be unique to the crystal structure of the targeted lattice. By combining de Broglie's theory with Bragg's law, a relationship can be derived between the diffraction angle of electrons and their accelerating voltage. This relationship can then be used to determine fundamental structural information about the crystal. Using this method, the inter-planar Spacing for d_{100} and d_{110} of graphite were determined to be 1.7 ± 0.1 Å and 1.4 ± 0.2 Å respectively.

I. INTRODUCTION

In the 18th century, Thomas Young conducted his infamous double slit experiment, challenging Newton's particle theory by proving the wave-like properties of light [?]. Later in the 19th century, Einstein published his work interpreting Philipp Lenard's experimental results on the photoelectric effect. In this work, Einstein concluded that light consists of quanta called photons, which act as particles [?]. This introduced the idea of wave-particle duality, the notion that these photons may behave as waves or particles depending on the situation. Building on Einstein's work, Louis de Broglie hypothesized that wave-particle duality may also hold for physical objects, not just for light [?].

In modern times, multiple experiments have shown the wave-like behavior of particles, but the first to prove de Broglie's hypothesis were electron diffraction experiments such as the Davisson-Germer experiment. The original intent of this experiment was to continue studying the angular distribution of electrons scattered by a nickel target. While collecting data, an accident occurred, requiring the target to be cleaned. After cleaning the target with high heat and effectively recrystallizing it, a possible relationship between scattering intensity and crystal direction was found, prompting further investigation. During this new investigation, they observed intensity peaks at specific angles, very similar to interference patterns and characteristic of wave-like behavior [?]. This was the first confirmation of de Broglie's theory. From there, known values of the target's interplanar spacing and diffraction angles were used to estimate the wavelength using Bragg's law. Those wavelength values were then compared to values calculated from de Broglie's formula. The results of that comparison were similar: calculations using both de Broglie's formula and Bragg's law yielded the same wavelength values. These results solidified de Broglie's theory and the idea of wave-particle duality for all matter.

II. THEORY

This experiment uses a very similar setup to the Davisson-Germer experiment and the same basic principles to work towards a slightly different goal. Instead of using measurements to estimate and verify wavelength calculations, this experiment aims to estimate the values of d_{100} and d_{110} for graphite. The Davisson-Germer experiment confirmed that de Broglie's formula can be used to accurately calculate the wavelength of an electron beam [?]. With this alternate method for determining wavelength and the ability to measure deflection angles, Bragg's law can be used to estimate d . Starting with de Broglie's formula

$$\lambda = \frac{h}{p}, \quad (1)$$

where h is Planck's constant and p is the momentum of the electrons as they are accelerated through a potential difference. This potential difference determines the speed that the electrons will travel at, and by translation, their momentum [?]. Bragg's condition states that

$$2d \sin \theta = \lambda n, \quad (2)$$

where d is the inter-planar spacing and λ is the wavelength, and θ is the angle between the target lattice and the incident beam, and n is the order of reflection[?]. In experiments like this one, it is common to simplify by assuming $n = 1$. This is because in electron diffraction, first-order reflections tend to be much more intense and visible than the higher-order ones [?]. If data is collected on the most visible reflections, then it likely corresponds to first-order reflections.

To better understand where θ comes from, refer to fig ?? above. The diagram shows an electron beam approaching the lattice at an angle θ with respect to the

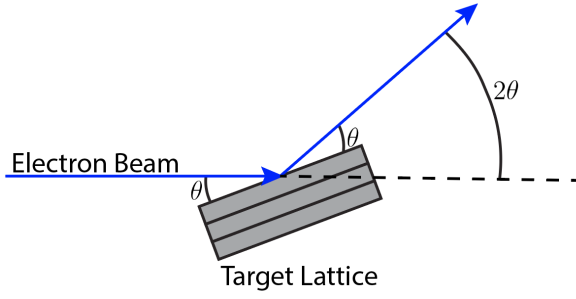


FIG. 1. Diagram for showing deflection angle for a beam of electrons bouncing off a target lattice.

lattice. Then it deflects off the lattice with the same relative angle. For the beam to approach and depart at an angle θ relative to the lattice, it must have deflected by a total of 2θ relative to the horizontal. This distinction becomes especially important when combining de Broglie's formula with Bragg's law to get

$$\sin \theta = \frac{h}{2d\sqrt{2m_e}} E^{-\frac{1}{2}}, \quad (3)$$

Where E is the energy of the diffracted beam given by $E = qV$. Equation ?? shows a linear relationship between $\sin \theta$ and $E^{-1/2}$. So if de Broglie was indeed correct about particles demonstrating wave-like behaviors, then a plot of measured values for $\sin \theta$ against the measured values for $V^{-1/2}$ should be linear just as the math suggests.

To test this and obtain real, measured values, an apparatus similar to the diagram shown in Fig. ?? is used. In this diagram, a beam of electrons is emitted from the filament, energized to the order of keV, and sent into the target lattice where it diffracts.

As seen in Fig. ??, the diffraction pattern forms a ring on the surface of the bulb. By measuring the distance a from the center of the ring to the ring itself, a value for $\sin \theta$ can be obtained.

$$\sin 2\theta = \frac{a}{L}. \quad (4)$$

Fig. ?? provides a more in-depth look at the geometry involved in obtaining Eq. ??.

L is the distance from the crystal lattice to the screen on the opposite end of the bulb. $L/2$ is roughly the radius of the bulb. The measured side length a completes an isosceles triangle. Equation ?? was derived by using the trigonometric ratio for $\sin$ on one of the two right triangles formed by an isosceles triangle. Given the trig identity $\sin 2\theta = 2 \sin \theta \cos \theta$ and the fact that for small angles $\cos \theta \approx 1$, Eq. ?? simplifies to

$$\sin \theta = \frac{a}{2L}, \quad (5)$$

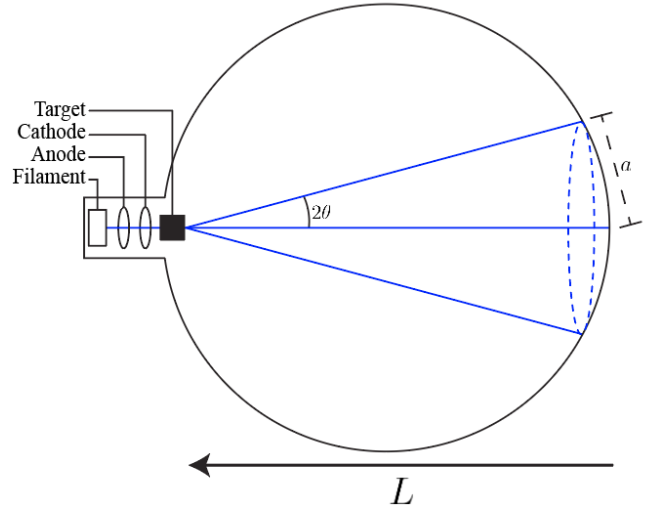


FIG. 2. A schematic representation of the diffraction tube on the 3B Scientific UE501050 electron diffraction apparatus.

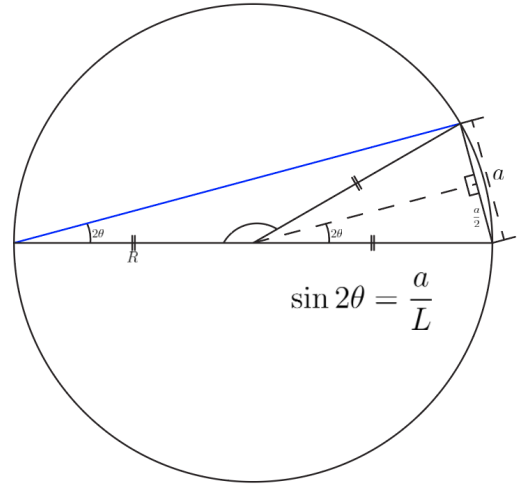
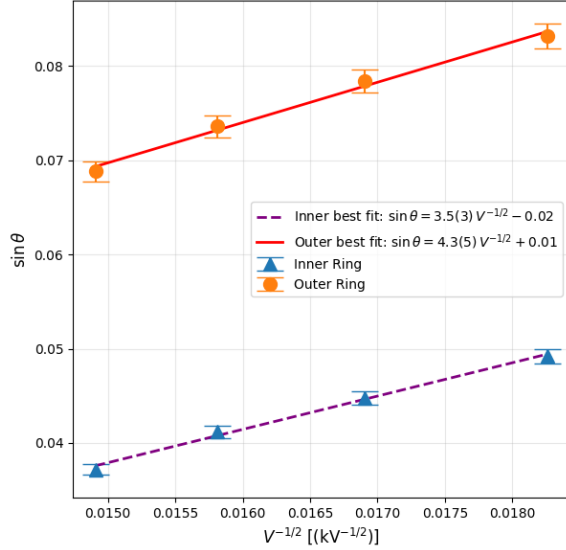


FIG. 3. A look at the geometry behind determining $\sin \theta$.

This equation provides a way to calculate values of $\sin \theta$ from the measured value of a and the L value provided in the manual. With that, all that remained was to take the physical measurements.

For data collection, voltage was varied between 3 and 4.5 kV. In this range, two rings formed on the bulb. These rings formed because the target lattice is a thin film consisting of graphite powder. The powder contains many small graphite crystals, so diffraction occurs at various orientations, resulting in ring formation. Measurements of a were taken for each ring at multiple points across the voltage range. The rings, while bright, were not sharp, so the measurements of a have an estimated uncertainty of 3 mm to account for the fuzziness of the

FIG. 4. Linear regression plot of $\sin \theta$ Vs. $V^{-1/2}$

rings. The other value, L , was not measured during the experiment but was given in the device manual. That value has an estimated uncertainty of 2 mm.

Fig. ?? is the plot resulting from the measurements of a taken from each ring at four different beam energies. For both rings $\sin \theta$ increases as $V^{-1/2}$ increases. This trend is consistent with the linear relationship $\sin \theta \propto E^{-1/2}$ shown by Eq. ?. The fact that the experimental results remain consistent with the earlier mathematical predictions confirms the accuracy of de Broglie's formula for this experiment.

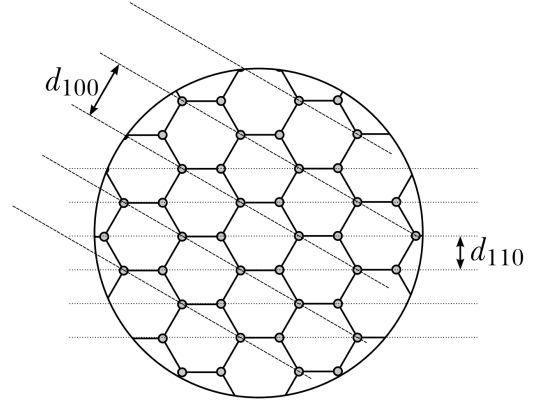
From here, the data can be used to calculate the d_{100} and d_{110} values of the crystal lattice using the slopes of the lines denoted by S . Solving Eq. ? for d and substituting slope S in place of $\sqrt{V} \sin \theta$ yields the equation

$$d = \frac{h}{2m\sqrt{2S_e q}}. \quad (6)$$

Plugging the slopes of the best-fit lines for each ring's measurements into equation ?? yielded an inner ring value of $1.7 \pm 0.1 \text{ \AA}$ and an outer ring value of $1.4 \pm 0.2 \text{ \AA}$. With two values for d , the next step is to figure out which value corresponds to which interplanar spacing. The answer becomes mathematically evident when combining Eqs. (3,5).

$$\frac{a}{2L} = \frac{h}{2d\sqrt{2m_e}} E^{-\frac{1}{2}}. \quad (7)$$

The resulting equation shows a to be inversely proportional to d , implying that ring projections with larger radii correspond to smaller interplanar spacings. A similar result is observed in the crystal structure of graphite itself.

FIG. 5. Visual of the graphite unit cell annotated with physical representations of the d_{100} and d_{110} spacings

Looking at Fig. ??, the spacing of d_{100} is shown to be physically larger than the spacing of d_{110} . This would imply that the inner ring value $1.7 \pm 0.1 \text{ \AA}$ corresponds to d_{100} as it is the larger of the two values, and vice versa for the outer ring value and d_{110} .

The last and final question concerns whether the values calculated from this experiment agree with the commonly accepted values for d_{100} and d_{110} . Other experiments find the value for d_{100} to be roughly $2.1319 \text{ \AA}$ and d_{110} to be rough $1.2308 \text{ \AA}$ [?]. Comparing results, the values for d_{100} are relatively far from each other while the values of d_{110} were much closer. The d_{110} value obtained from other sources is actually contained within the estimated uncertainty of this experiment's d_{110} value, while d_{100} is not even close.

III. CONCLUSION

This experiment used an apparatus similar to that used in the Davisson-Germer experiment to revisit the idea of wave-particle duality in physical matter, leveraging it to perform basic analysis of the interplanar spacing of graphite. By combining Bragg's law with de Broglie, it was determined that $\sin \theta \propto E^{-1/2}$. When experimental results demonstrated trends consistent with that linear relationship, the data was used to calculate values for interplanar spacing. Of the two values $d_{100} = 1.7 \pm 0.1 \text{ \AA}$ and $d_{110} = 1.4 \pm 0.2 \text{ \AA}$, only d_{110} was close to widely accepted values of its interplanar spacing. In either case, the difference likely stems from measurement error during this experiment. Using manual measurement tools on a circular bulb with fuzzy reflections is not the most accurate way of going about this kind of analysis. Even with inaccurate values, this experiment was successful in its demonstration of de Broglie's theory and the wave-like nature of matter.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]]

Estimating Planck's constant using the photoelectric effect

Sema'Je Zechariah Farmer

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

The photoelectric effect occurs when light of a sufficiently high frequency strikes a surface, exciting the electrons on that surface to the point that they are ejected from it. Those ejected electrons are called photoelectrons, which can be collected and sent through a circuit. By measuring the resulting current and the negative voltage required to stop it at various frequencies and intensities, some essential relationships can be established. The first is the relationship between stopping voltage and frequency. This relationship is linear, and using its slope, Planck's constant was estimated to be $6.5(2) \times 10^{-34}$ Js. The second relationship is between the stopping voltage and intensity. Classical wave theory predicts that an increase in intensity should increase the stopping voltage. In reality, the data shows that the two are entirely unrelated. This result directly contradicts wave theory, supporting theories of light's particle-like behavior and the existence of photons.

I. INTRODUCTION

The debate on whether light behaves as waves or particles spanned multiple centuries. It began in the early 1600s with various theories on both sides of the argument, but concluded in the early 1900s with Einstein's explanation of the photoelectric effect [?]. Heinrich Hertz originally discovered this effect [?], and later Philipp Lenard studied it [?]. Lenard used a carbon-arc light to excite electrons on a metal plate. Those excited electrons would then be ejected from that plate and collected on another plate located some distance away from the emitter. The collector plate was then connected to a cathode and a sensitive ammeter, allowing him to measure the current induced by the collected electrons. In his studies, Lenard compared the energy of the photoelectrons (ejected electrons) to the intensity of the light coming from the source. To measure the energy of the electrons, Lenard applied a negative charge to the collector plate, repelling any electrons that lacked sufficient energy to pass. In doing so, Lenard discovered a quantity called the stopping voltage, the voltage applied to the collector plate that prevents all electrons from reaching it. Through testing, Lenard found that this stopping voltage remained constant regardless of the source's light intensity [?]. He even went so far as to test this with different colors, finding that the stopping voltage actually depends on the wavelength of light [?]. Einstein suggested that light radiation should be thought of as quanta called photons, each with energy hf , where each photon transfers its energy to a single electron.

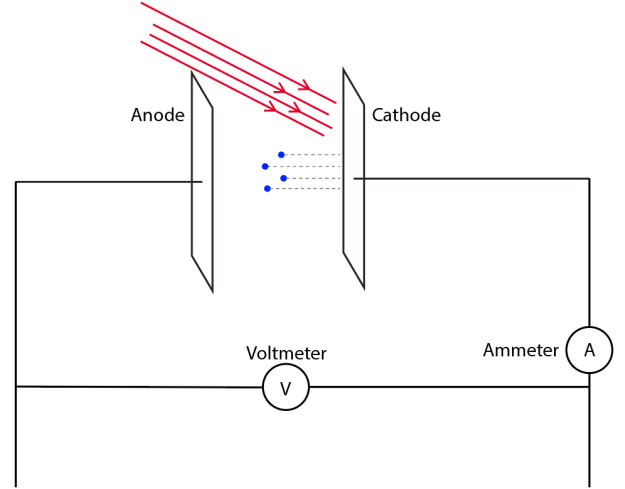


FIG. 1. Basic circuit of the photoelectric effect. Light strikes the emitter plate on the right, exciting the electrons enough to eject them towards the collector plate on the left. An ammeter is included in the circuit to measure the induced current, and a voltmeter is included to measure the potential difference between the two plates.

II. THEORY

In this experiment, we used the PASCO SE-6609 photoelectric effect apparatus to recreate Lenard's experiment. The apparatus uses a mercury lamp to illuminate a receiver. The receiver then filters out all but one wavelength of light. From there, the filtered light hits the emitter plate, ejecting electrons into the collector plate, and we measure the current induced by those electrons as seen in fig ??.

Similar to Lenard's experiment, we measure the cur-

rent induced at different intensities and the stopping voltage at various frequencies. We then attempt to use that collected data to estimate Planck's constant and confirm Einstein's photon theory.

If the energy (E) of a photon is equal to Planck's constant (h) multiplied by the photon frequency (f), and that photon only transfers its energy into a single electron [?], then the energy of the ejected electron can be described by

$$E = hf - \phi, \quad (1)$$

where ϕ is the work function for the cathode. The work function represents the amount of work required to free electrons from a given material and, as such, is dependent on the material surface of the cathode. In the case of the PASCO apparatus used in this experiment, the material is likely a light-sensitive metal with a low work function. The stopping voltage (V_{stop}) can be derived from eq. ?? by substituting in eV for the energy and solving for V .

$$V_{stop} = \frac{h}{e}f - \frac{\phi}{e} \quad (2)$$

where e is the charge of electrons. This equation shows a linear relationship between V_{stop} and f . So a plot of V_{stop} vs. f should result in a slope of h/e and a y-intercept of ϕ/e . The goal of this experiment is to demonstrate the relationship with physical data that can then be used to estimate Planck's constant.

To do so, we collected two datasets. The first was stopping voltage vs intensity. This dataset was collected to verify that the stopping voltage is independent of light intensity. The stopping voltage was recorded by increasing the negative voltage applied until the current read zero. The lowest voltage magnitude at which the current stopped was recorded as the stopping voltage for that test. Intensity was not directly measured; instead, it was implied. It is well documented that the intensity of electromagnetic radiation decreases with increasing distance from the source to the receiver. This relationship is known as the inverse-square law [?], which states that $I \propto r^{-2}$ [?]. So changing r (the distance) will inversely affect I by a factor of r^2 . With this in mind, the stopping voltage was measured at four different distances (and by translation intensities) for four different wavelengths. The second data set was the stopping voltage vs frequency. This was collected to both establish that stopping voltage depends on frequency and to estimate Planck's constant. The stopping voltage we measured using the same methodology as the other data set, and the frequency is given by the selected wavelength.

Measurements of stopping voltage vs intensity yielded the expected results. It was hypothesized that the stopping voltage at a given frequency should remain constant regardless of light intensity. So we'd expect V_{stop} to stay the same for any distance.

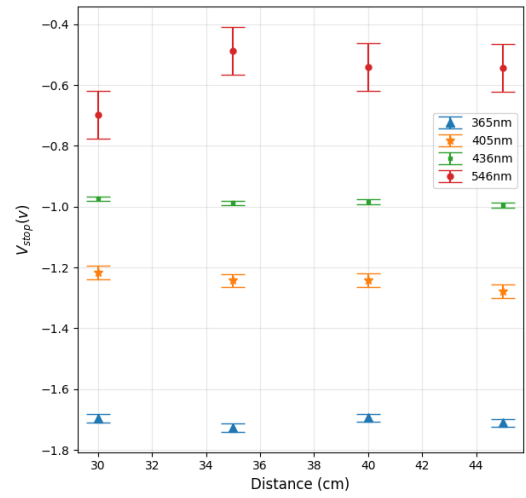


FIG. 2. Stopping voltage versus distance scatter plot. Measuring the stopping voltage at various distances for four different wavelengths of light.

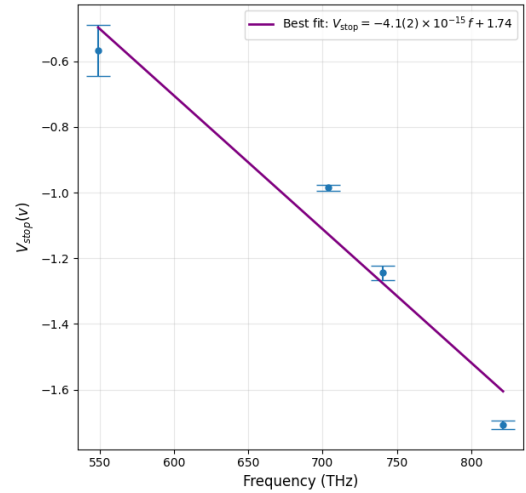


FIG. 3. Stopping voltage vs frequency scatter plot with slope h/e .

Fig. ?? shows the results of data collection for all four wavelengths. For all four wavelengths, a hypothesis test was conducted to determine how far each V_{stop} is from the mean in its series. The result of this analysis was good across the board, with Z scores across all series remaining just at or below ± 0.75 . This indicates a high level of consistency and no significant deviation from the mean. The fact that the measurements taken for the stopping voltage at each wavelength remain mostly constant shows that the stopping voltage is not dependent on the intensity of light. This result directly contradicts the wave theory and supports the particle theory of light.

Data collection and analysis on stopping voltage vs. frequency was just as successful. Based on eq.?? we expect physical results to show a linear relationship be-

tween stopping voltage and frequency.

Fig. ?? shows the expected result: the stopping voltage decreases as the frequency increases. By rearranging eq. ??, we find that Planck's constant (h) should equal the slope of the line (V/f) multiplied by the charge of the electrons (e). Similarly, the work function for the cathode can be obtained by multiplying the y-intercept by e . Using these methods, Planck's constant was estimated to be $6.5(2) \times 10^{-34}$ J s, and the work function to be 2.78 eV. The accepted value for h as reported by NIST is 6.626×10^{-34} J s [?], leaving a percent difference of only 1.92% between the two values. Most common work functions generally fall between the values of 2 eV and 5 eV [?]. So while we don't know the exact work function of this apparatus's cathode, the experimental value still falls within the typical range.

For both data sets, the primary source of error was the voltage measurements. During data collection, the surrounding area was not entirely dark. This made it challenging to determine the exact cutoff for the stopping voltage, as ambient light induced a small but constant current. This problem became especially difficult for higher wavelengths such as 546 nm, and 577 nm (not included), where the current induced by the light source was not much larger than that of the ambient light. As a result, the current never truly reaches 0 when the voltage is applied, leading to a larger margin of error. Lower frequencies most often induced a current value that was an order of magnitude higher than the ambient light. On that larger scale, the contributions of ambient light weren't nearly as noticeable. This would be why the uncertainty for wavelength 546 nm is so much larger than the rest.

III. CONCLUSION

This experiment aimed to demonstrate and confirm Einstein's theories about light using the photoelectric effect. By measuring the stopping voltage at various frequencies and light intensities, we established that, in the photoelectric effect, light behaves as particles. Some of the evidence gathered for this was that the stopping voltage depended on frequency and not intensity. This implies that the interactions between light and electrons are more individualized, with electrons absorbing only as much energy as a single photon can transfer. Einstein's theory was then further confirmed by the fact that Planck's constant could be calculated within 2% of its actual value using equations consistent with particle theory. This work does not disprove wave theory as a whole but instead introduces the idea of wave-particle duality, in which the behavior of light is context-dependent. In some cases, like the double slit experiment, the wave model clearly makes the most sense. At the same time, other experiments, like the photoelectric effect, can only be explained by particle theory.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

Dispersing Doubt: how wavelength affects refraction.

Skylar Farr

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

A comparison of the index of refraction of two lasers as refracted through the same media will result in the dispersion. By using the Michelson Interferometer experiment to find the wavelengths of various lasers and a semi-cylindrical prism to calculate the indices, the different of the two indices can be taken to get the dispersion of the two different wavelengths from the lasers. The calculated dispersion was 0.014 ± 0.13 . This dispersion is consistent with what would be expected from only two separate lasers that were not many wavelengths apart. Given more options for laser colors to analyze, the data would be a lot more accurate.

I. INTRODUCTION

Indices of refraction, interferometry, wavelength, dispersion. All terms that are encompassed by the field of optics. Each has a rich history and background rooted in curiosity to understand the world around us. Starting with interferometry which was used to find the wavelengths of light and other forms of waves [?]. Albert A. Michelson, an American physicist, invented the Michelson Interferometer in order to prove the existence of the aether. He ended up with null results, but post this invention, it has been used consistently to measure wavelengths of light and, in the case of this experiment, lasers. Different types of lasers have different wavelengths, this fact does not affect the interferometry process but it does affect indices of refraction. An index of refraction is the ratio of the speed of light in two different media and it determines how much that light is bent or refracted when entering a material [?]. For differences in wavelength, despite having consistent media when measuring the values, the indices will vary. This variation is called dispersion. Dispersion is the scattering of light away from its central point or into multiple different components. Dispersion is relatively simple to understand but it can be used in many complicated systems and it is important for understanding the behavior of different waves. It is very important for types of imaging to ensure that light does not bleed and that the image being displayed is sharp and clear. Additionally, it can be used in fiber-optic communication systems.

II. RED VS. GREEN

Observing dispersion through a medium depends on two things, the wavelength of the light or laser being used, and the index of refraction of the medium. Both of these variables can be measured experimentally. To determine the wavelength, an experiment can be per-

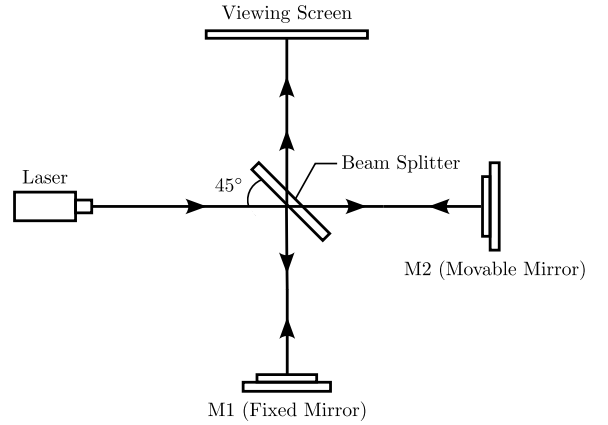


FIG. 1. Schematic of a Michelson Interferometer. A laser is shone into a beam splitter angled at 45° where it is reflected off of a fixed mirror $M1$ and a movable mirror $M2$, onto a viewing screen where the fringes of the wave are visible.

formed using a Michelson Interferometer as seen in Fig. ?? . The equipment used for this experiment was the PASCO Model OS-8501 Interferometer and it works by splitting a laser beam and reflecting it off of a stationary and movable mirror onto a viewing screen where the fringes of the wave are visible, as seen in Fig. ?? . From here, the movable mirror can be adjusted in micron graduations which causes the fringes on the viewing screen to change. By measuring the distance the mirror is moved, and comparing it to the number of fringes that pass on the viewing screen, the wave length can be calculated using this equation,

$$\lambda = \frac{2d}{m}, \quad (1)$$

where λ is the wavelength, d is the distance $M2$ is moved, and m is the number of fringes counted.

Once the wavelengths are calculated the indices of refraction can be calculated. This is done by using Snell's

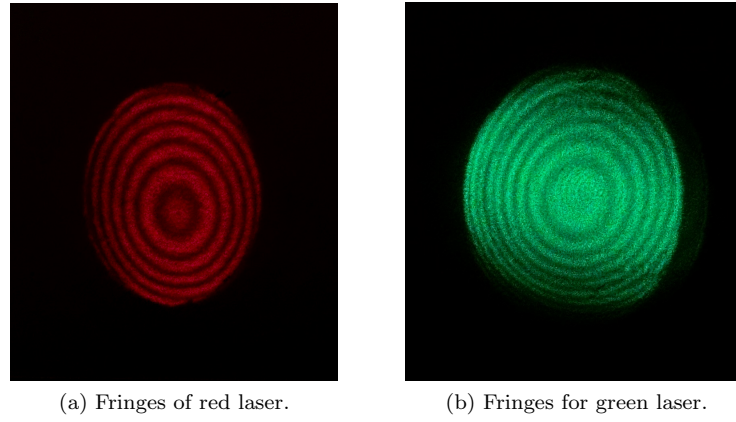


FIG. 2. The resulting projection of the laser after it has gone through the Michelson Interferometer.

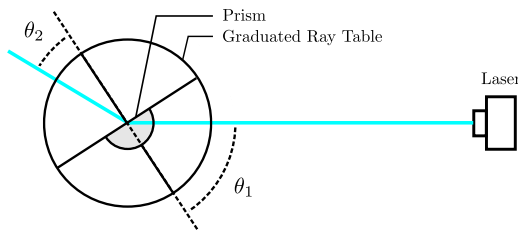


FIG. 3. Schematic of the index of refraction experiment. The laser is directed into the semi-cylindrical prism placed on top of the graduated ray table. The ray table is rotated to the angle of incidence θ_1 which produces the outgoing angle θ_2 .

Law to find the index of refraction of the medium the lasers are passing through. Snell's Law is

$$n_1 \sin(\theta_1) = n_2 \sin(\theta_2), \quad (2)$$

where n_1 is the index of refraction of the first medium the laser passes through, which in this case is air at a value of $n_1 \approx 1$, θ_1 is the angle of incidence, or the angle that the laser enters the second medium at, θ_2 is the outgoing angle, and n_2 is the index of refraction of the second medium which is the value that is in the process of being calculated. This experimental set up contains a graduated ray table, and a semi-cylindrical acrylic or PMMA plastic prism as seen in Fig. ??.

For both of the separate experiments, two sets of data were taken; once for a red laser and once for a green laser. The red laser is a PASCO OS-8525A Basic Optics Red Diode Laser with a wavelength $\lambda_r = 650$ nm, and the green laser is a PASCO OS-8458B Basic Optics Green Diode Laser with a wavelength of $\lambda_g = 515$ nm. After both sets of data were taken, each experiment went through specified linear regressions and error analysis. The estimated uncertainty for the wavelength of both lasers was $\delta = \pm 1$ μ m. The estimated uncertainty for the index of the red laser was $\delta_r = \pm 3^\circ$ and the green laser was $\delta_g = \pm 2^\circ$. These uncertainties were used in the error propagation of the data sets.

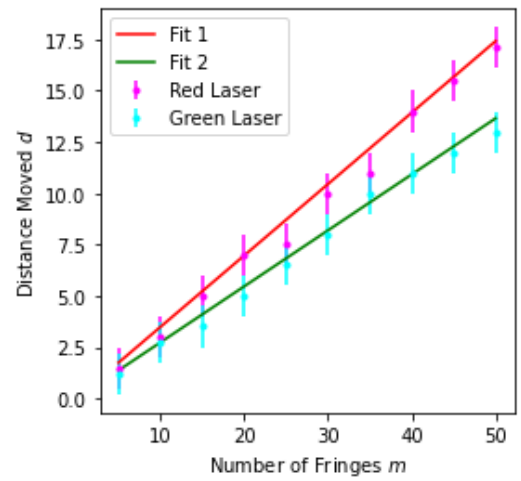


FIG. 4. Number of fringes vs. distance moved. Graph of the slope of each laser after a linear regression to determine the wavelength. Error bars are based off of the estimated uncertainty.

Starting with wavelength. The calculated wavelength based on the slope of the plotted data in Fig. ??, is $\lambda_r = 697 \pm 14$ nm for the red laser, while green is $\lambda_g = 546 \pm 14$ nm. These values for wave length are consistent in showing the behavior of a wave when it is measured by a Michelson Interferometer, but the data is not accurate when compared to the measured wavelengths of both lasers. This is a little bit surprising but not completely unexpected. When the knob that controlled the movement of M2 was rotated it would often continue to move slightly in the opposite direction that it had been spun once it was let go since it was spring loaded. This cause a few instances of having to completely restart during the counting of the number of fringes or it moved without being noticed and the fringes were miscounted.

Now that the wavelength data has been analyzed and the wavelengths have been found, the index of refraction can be found. The purpose of finding the wavelength be-

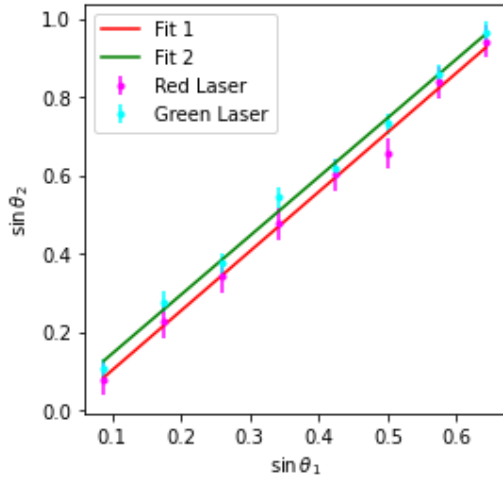


FIG. 5. Graph of $\sin(\theta_1)$ vs $\sin(\theta_2)$, where the slope of each laser is plotted with a best fit after linear regression to determine the index of refraction. The error bars are based off of the estimated uncertainty.

fore the index of refraction was to understand the difference in properties of the two lasers, under the assumption the recorded values were not available. The calculated indices based on the slope of the data plotted in Fig. ??, is $n_r = 1.52 \pm 0.08$ for the red laser, and $n_g = 1.51 \pm 0.05$ for the green laser. These values are relatively close to the recorded values of the index of refraction for the respective lasers [?]. Despite any errors, these data sets successfully show the expected behavior of the index of refraction through acrylic.

The last, but most important step, is calculating the dispersion. This is done by taking the difference of two indices, and calculating the uncertainty from the uncertainties of the two indices as seen below

$$\Delta n = n_r - n_g, \quad \delta_n = \delta_r + \delta_g, \quad (3)$$

which for this data gives, $\Delta n = 0.014 \pm 0.13$. While this may look like an extremely bad number due to the uncertainty being larger than the calculated value, its actually pretty normal. Since the dispersion is being found with only two wavelengths that are already close in number, it is not surprising that the dispersion is so small. Since you can only add uncertainties not subtract despite the original equation needing subtraction, it just grows.

III. CONCLUSION

As it has been found, dispersion can often be an incredibly small number with a high uncertainty which makes precision important whenever it comes to using dispersion in any application. It is very simple to find, as long as you have all the right components, but taking a measurement that is truly accurate and precise takes a lot of time and patience. Within both experiments, there

were a lot of opportunities for error and as much of that error was taken into account as possible. Comparatively, the data here is not very accurate because of the difference between the recorded and measured values, but the behaviors of the data aligned with what was expected. This kind of error can be systemic, there could be something wrong with the equipment being used. The age of the equipment could have caused it to degrade over time. Another systemic error could be a problem with the person taking the data. It is possible that the data was consistently misread the same way every time data was recorded.

Again, regardless of any inaccuracy, this experiment was still able to take somewhat reasonable numbers and calculate a dispersion between a red and green laser as it is seen through an acrylic prism. The experiment has really reflected how much every part of optics feeds into each other and any doubts that previously plagued the minds of readers is, ideally, dispersed.

ACKNOWLEDGMENTS

This research would not have been made possible without funding from the Carthage College Physics and Astronomy Department along with assistance from Dr. Joseph Anderson.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]

Reflecting on microwaves with Lloyd's mirror

Skylar Farr

*Department of Physics and Astronomy, Carthage College,
2001 Alford Park Drive, Kenosha, WI 53140, USA*

(Received January 29, 2026, Published online February 5, 2026)

The wavelength of a microwave transmitter was determined by using a method of interferometry called Lloyd's Mirror. This method consists of a transmitter, reflector, and a receiver. By relating the total path a wave travels, to the distance of a reflector away from the wave, the wavelength of said microwave can be calculated using a linear regression. Two sets of data taken are consistent in showing distance does not change wavelength despite the imprecision of the results when compared to the recorded wavelength of the microwave transmitter. Outside influences creating additional reflection are likely the cause of this imprecision.

I. INTRODUCTION

Interferometry is a term used to describe the process in which the interference of waves are measured in order to determine various forms of data. This process has been used for many years to detect different variables and physical interactions. A popular example is from American Physicist Albert A. Michelson who created the Michelson Interferometer to measure the effects of Earth's movement on the speed of light. This experiment, fueled by curiosity, ended up showing the lack of existence of what Michelson thought was a medium surrounding the Earth, known as the aether [?]. The Michelson interferometer is not the only use of this technology; it is also used in something called Lloyd's Mirror. This technique was made by Irish physicist Humphery Lloyd to learn more about the wave nature of light. However, unlike the original experiment, the specific set up for this experiment uses electromagnetic waves, EM waves, rather than light [?]. Despite the difference in the type of waves, this method works by measuring the interference of waves in order to determine constants, such as the wavelength of the wave being used.

Today, some other more common uses of interferometry and, more specifically, Lloyd's mirror are optical measurement, radio astronomy, and underwater acoustics. All of these applications put into use the different aspects of Lloyd's mirror, like understanding reflection and refraction of waves, being able to apply transformations to waves, and understanding constructive and destructive interference.

II. THE EXPERIMENT

Now that the some general information on interferometry has been laid out, it's time to understand Lloyd's Mirror more in depth. As stated above, it works by measuring the interference of waves. There are different inter-

actions that happen between the waves to create interference. The first kind being constructive interference. This is when two waves align in the same phase to combine and become amplified. The second type is destructive interference where the two waves become out of phase and begin to cancel each other out. If two waves are 180° out of phase, it is possible for them to completely cancel each other out. For the purposes of this lab, the waves were measured when they were in a state of constructive interference, or at the peak of their wavelength. The purpose for measuring these waves is to determine the wavelength based on the intensity of said waves which is determined by how their constructive and destructive interference changes.

The set up of this experiment consists of the PASCO Microwave Transmitter WA-9801 and Microwave Receiver WA-9800, and other elements of the PASCO Microwave Optics System (WA-9314C) including component holders, metal reflectors, the fixed arm assembly, and the goniometer. The receiver is equipped with a current sensor with a multiplier and sensitivity dial to be able to read the intensity of the waves it receives. For this experiment this display was simply used as a reference to be able to note the constructive and destructive interference of the waves. Both the transmitter and receiver are positioned along a meter stick with the goniometer in the center to allow for ease in adjustment of the angles of the transmitter and receiver. Additionally, the fixed arm assembly is placed onto the degree plate to be able to position the metal reflector plate. When all the equipment is set up, the transmitter and receiver should be directly across from each other with the openings facing each other and the reflector plate should be slightly offset, facing inward, while positioned at an equidistant point between the two, as seen in Figure ??.

The values being measured are the distance from the transmitter to the center of the goniometer d , and the distance of the reflector from the center point of the goniometer to the metal reflector h as its distance changes

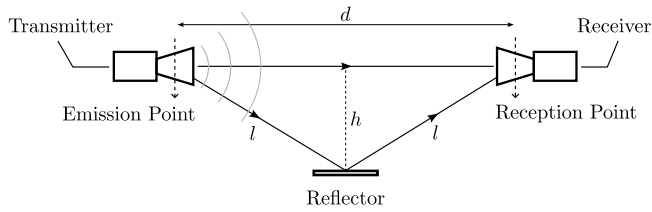


FIG. 1. A schematic of the experimental setup containing an EM wave transmitter and receiver set up at distance d apart, and a reflector plate at distance h from the equidistant point between the two devices. The total path traveled by the wave is noted by l .

with respect to the changing intensity of the waves. The values of eight different maxima of the wave intensity was measured at two different distances d . The equation to determine the wavelength from these values is

$$2l - d = n\lambda, \quad (1)$$

where $2l$ is the total path traveled by the wave, n is the number of waves or "fringes" that have been measured out from the first maximum point, and λ is the wavelength. To be able to solve the equation, the value $2l$ must be found first. This can be done by using the Pythagorean theorem,

$$a^2 + b^2 = c^2, \quad c = \sqrt{a^2 + b^2}. \quad (2)$$

After obtaining the value of $2l$ a linear regression can be performed on the data to plot the number of fringes vs. the total length traveled where the slope is the wavelength. This fit along with error bars is seen in figure ???. The estimated uncertainty of the distance of the reflector from the goniometer h is ± 0.5 cm. When adjusting that distance, the dial displaying the current was not incredibly sensitive, so slight adjustments of the reflector often made little to no difference in the maximum intensity being displayed. This caused a lot of back and forth movement of the reflector to get a best estimate for the peak of the intensity and that back and forth movement often varied by about 0.5 cm.

As mentioned above, there were two sets of data taken, one where the distance between the transmitter and receiver was $d_1 = 100$ cm, and the other where $d_2 = 80$ cm. The goal with taking two sets of data was to see how distance affects the wavelength, while knowing that it should not. After plotting both data sets, the measured wavelengths for d_1 and d_2 are $\lambda_1 = 2.68$ cm and $\lambda_2 = 2.52$ cm. Taking into account the estimated uncertainty ± 0.5 cm, which is the same for both data sets, and performing an error analysis, the final uncertainty of the wavelengths is, $\delta = \pm 0.08$ cm, making the final values $\lambda_1 = 2.68 \pm 0.08$ cm and $\lambda_2 = 2.52 \pm 0.08$ cm. The recorded wavelength of the microwaves emitted from the PASCO transmitter is $\lambda = 2.85$ cm [?].

Although these data are consistent within the data sets and despite the difference in distances between the trans-

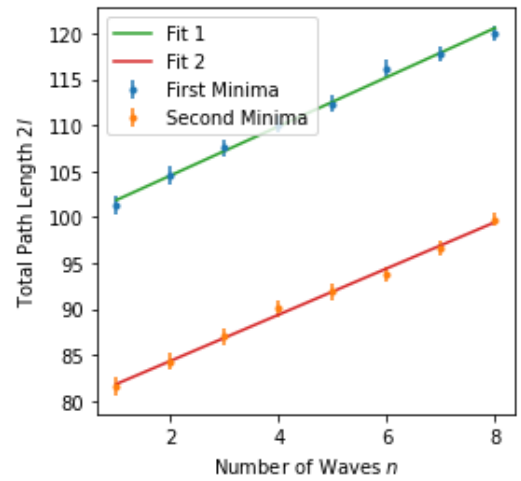


FIG. 2. Number of fringes vs. total path length the wave traveled. The slope of the fit line represents the wavelength of the microwaves emitted by the transmitter.

mitter and the receiver, figure ?? still shows a linear relationship between the two data sets, the precision of the data compared to the recorded value of PASCO is significantly different. This is likely due to external interference and reflection from objects other than the receiver. If the person recording the data and moving the reflector recorded the maxima while positioned near or behind the reflector, that person would also act as a reflector. If the experiment was set up on an inconsistent surface, the adjustment of the reflector could have been inaccurate. This leads to the belief that the significant difference in the recorded vs. measured values has to do with some external factor since the results are consistent.

III. CONCLUSION

This experiment was both successful and unsuccessful. The data gathered was consistent in displaying the linear relationship between distance the wave traveled and the number of fringes passed, and the results of the slope for both data sets accurately displayed the data, but the data was still imprecise to the recorded values of the transmitter. As mentioned above, this was likely caused by the person recording the data creating additional reflections or the condition of whatever surface the experiment was performed on. When performing data analysis, this potential error was not taken into consideration as the conditions in which all of the data was taken was consistent, meaning the data would be accurate within the sets taken and the amount of additional reflection would be the same for both data sets.

In the future if this experiment were to be replicated, an important recommendation would be to find a smooth surface, along with finding a way to reduce any additional reflection. This could mean moving the reflector

very slightly and backing away from the experimental set up to observe and record the status of the intensity, rather than remaining directly next to the equipment. This process may be more tedious as it will take many more iterations of adjustment, but the results would be more precise. Again, despite being imprecise, the data is still accurate and it correctly displays the relationship between distance and wavelength.

IV. ACKNOWLEDGMENTS

This lab was completed with the collaboration from lab partner Brookelyn Vermont. Additionally, this research would not have been made possible without funding from the Carthage College Physics and Astronomy Department along with assistance from Dr. Joseph Anderson.

DATA AVAILABILITY STATEMENT

A Jupyter notebook containing all the experimental data, data analysis, figure generation, and additional information on uncertainty analysis can be found in the supplemental materials. [?]